

Yichen Kong

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High School Researcher | Focus: Aerospace Design, Automation Algorithms, RF Comms



Profile Summary

Motivated by the high operational costs and marine ecological pollution of traditional disposable radiosondes, I engineered an autonomous, recoverable sounding system. By integrating morphing aircraft structures, low-power RF communications, nonlinear fully-coupled control, and Model-Based Design (MBD), this system offers a high-precision, eco-friendly, and cost-effective alternative for global high-altitude atmospheric data collection.

Core Research Project

- Intelligent Autonomous Recoverable Sounding System
- Oct 2024 – Present
- Architecture: Integrates a morphing flight payload, MAVLink ground station, and web-based terminal. Lifted by a 750g balloon to 30,000m, the payload separates and executes an unpowered glide recovery, resolving uncontrollable descents, funding drains, and data limitations.*
- Aerodynamics:** Designed a “Morphing Swept Wing + Missile Fin” hybrid configuration. Dynamic wing-sweep adjustments improve wind resistance; all-moving straked fins enable precise attitude control without conventional surfaces. Integrated a von Kármán nose cone and hybrid wingtips.
 - Parametric Modeling & CFD:** Built 3D models (SolidWorks) and executed multi-scale CFD simulations (XFlow, XFLR5, ANSYS-Fluent, STAR-CCM+) to validate cross-Reynolds-number performance and asymmetric flow characteristics.
 - Hardware Engineering:** Engineered an automotive-grade 6-layer high-frequency PCB (JLCPCB EDA), incorporating a synchronous buck-boost converter, X7R capacitors, and thin-film resistors for extreme thermal stability.
 - Ground Station & Terminals:** Developed a custom MAVLink ground station. Built a web terminal embedding ISA atmospheric models and the WMO-recommended Dines formula (solved via bisection) to simulate real-world sounding parameters.
 - Extreme-Environment Manufacturing:** Achieved a high strength-to-weight ratio using CNC-machined PMI foam (75kg/m³), FDM-printed HT-PLA-GF, and MJF-printed PA11 nylon. Applied DIN-compliant double-layer sealing and specialized lubricants for stratospheric reliability.
 - Flight Control Algorithms:** Implemented flight control via MBD. Applied auto-tuning PID and Total Energy Control Systems (TECS) to reject nonlinear perturbations during balloon separation and stabilize long-distance return glides.
 - Long-Range RF Link:** Designed a custom UHF RF link with LoRa CSS modulation and a 12-element Yagi antenna. Achieved robust bidirectional transmission at power levels significantly lower than industry standards (e.g., Vaisala radiosondes), eliminating signal dropout.

Flight Testing & Compliance

- Tuotuo River National High-Altitude Meteorological Station & Sanjiangyuan
- Jul 19 – 22, 2026
- Validation:** Directed extreme-environment tests at 4,534m altitude (severe cold, low pressure, turbulence). Secured rigorous airspace approvals from CAS institutes and regional met bureaus. As the sole flight test director, I successfully oversaw drone-assisted low-altitude drops and 30,000m stratospheric balloon launches in strict adherence to national regulations, validating core thermal and dynamic resilience.

Achievements & Honors

1. 40th National Youth Science and Technology Innovation Competition
- Most-awarded contestant across all categories
- Top honor:** CAS President’s Award
 - 7 special awards:** Gekko Special Award, Zhou Peiyuan Science and Technology Innovation Award, Mao Yisheng Science and Technology Award for National Youth Science and Technology Innovation, Tunisia Future Science and Technology Association Excellence Award, “Future Star” Youth Automotive Engineering Inspiration Award, “Aerospace Youth” Aviation Science and Technology Innovation Dream-Building Award, Jianda Zhixing Award
2. National Invention Patent (Sole Inventor)
- App No.: 202610625965.2
- Status:** Passed preliminary examination by CNIPA; currently under substantive examination
3. National Youth Science and Technology Innovation Competition
- Two-time Finalist
4. Qinghai Youth Science and Technology Innovation Competition
- Two-time First Prize
- Declaration:** All aerodynamic designs, control algorithms, and system assembly were completed independently
5. CMA Certification:
- Self-designed radio transceivers passed testing by the Qinghai Radio Testing Station
- Testing body:** Qinghai Radio Testing Station
6. Empirical Validation:
- Custom sounding calculation terminal passed operational field testing

Technical Skills

- Certifications:** Amateur Radio Operator Certificate (No. A6300000049); holds a PRC Radio Station License.
- Engineering:** Parametric modeling, Multiphysics CFD, 6-layer high-frequency PCB, RF comms, SDR, Front-end development, STM32 embedded architecture, MBD algorithms, and automation control strategies.

Expert Endorsements

Received technical validation and support from leading experts: Senior Eng. W. Huang (Aerospace Information Research Institute, CAS), Prof. J. Bian (UCAS), Senior Eng. Z. Bai (Institute of Atmospheric Physics, CAS), Prof. D. Zhang (SCUT), Assoc. Prof. Z. Du (Qinghai Univ.), and Director H. Guo (Xining National High-Altitude Met. Station).